College of Vocational studies

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Machine Learning Practical File

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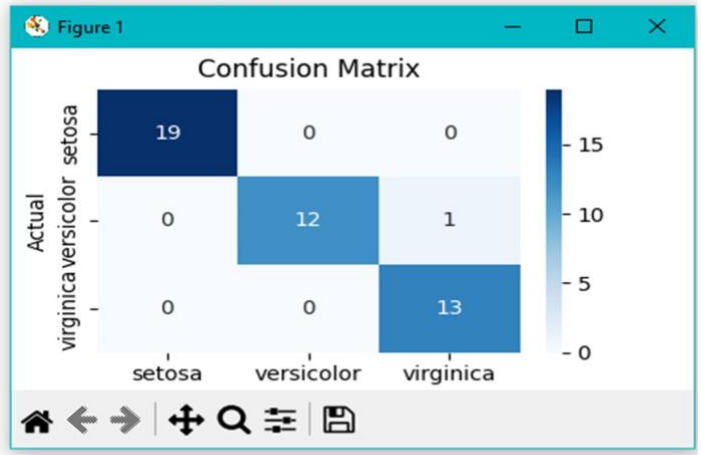
Semester - VI

1 Naïve Bayes Classifier

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
data = load_iris()
X, y = data.data, data.target

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Initialize and train model
model = GaussianNB()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate
print(f'Accuracy: {accuracy_score(y_test, y_pred) * 100:.2f}%')
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=data.target_names))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=data.target_names,
yticklabels=data.target_names)
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```

Output:

```
== RESTART: C:\Users\user\Documents\machine learning\1NaiveBayesClassifier.py ==
Accuracy: 97.78%

Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        19
  versicolor       1.00      0.92      0.96        13
   virginica       0.93      1.00      0.96        13

    accuracy                           0.98        45
   macro avg       0.98      0.97      0.97        45
weighted avg       0.98      0.98      0.98        45
```

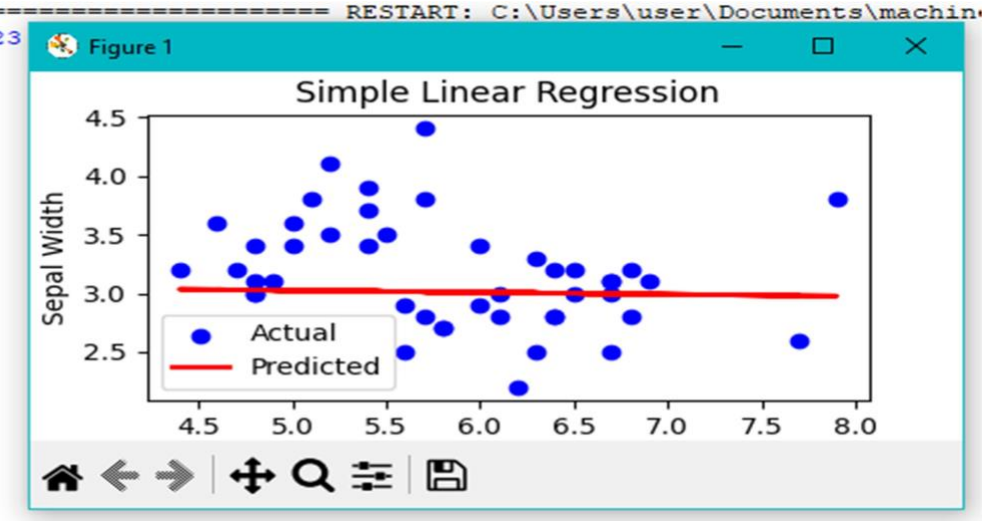


2 Simple Linear and Multiple Linear Regression

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
import matplotlib.pyplot as plt

# Load dataset
data = load_iris()
X = data.data[:, 0].reshape(-1, 1)  # Using sepal length as feature
y = data.data[:, 1]  # Using sepal width as target

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Initialize and train model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate
print(f'Mean Squared Error: {mean_squared_error(y_test, y_pred):.2f}')
print(f'R^2 Score: {r2_score(y_test, y_pred):.2f}')

# Plot
plt.scatter(X_test, y_test, color='blue', label='Actual')
plt.plot(X_test, y_pred, color='red', linewidth=2, label='Predicted')
plt.xlabel('Sepal Length')
plt.ylabel('Sepal Width')
plt.title('Simple Linear Regression')
plt.legend()
plt.show()
```

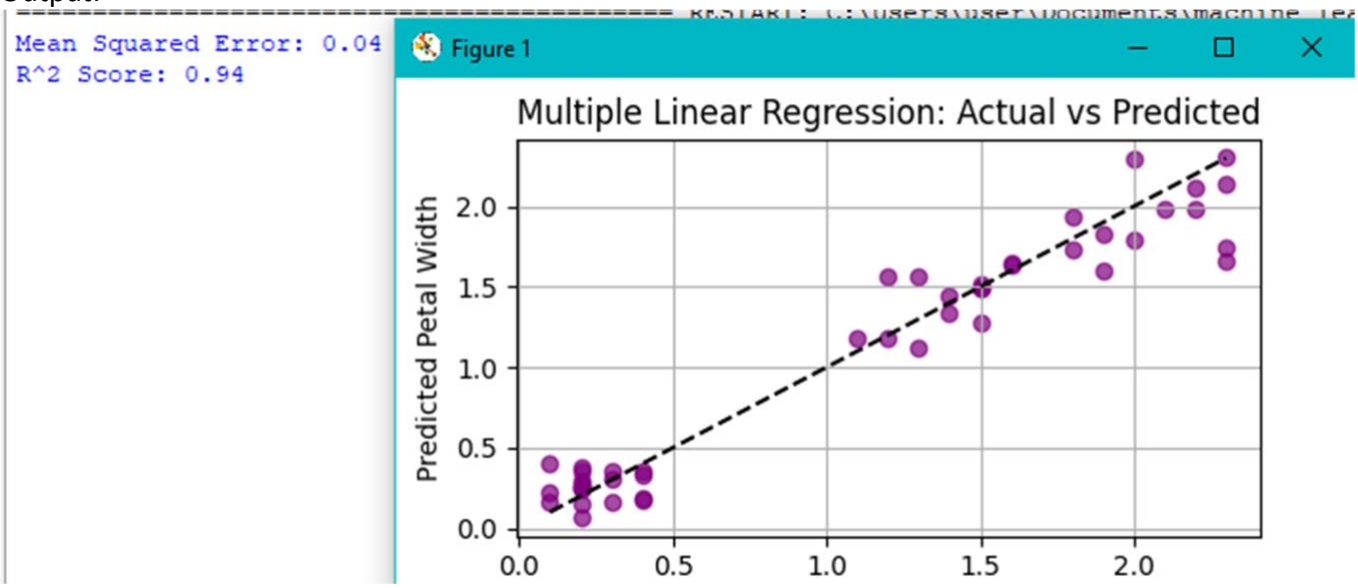
Output:

```
========================================== RESTART: C:\Users\user\Documents\machin
Mean Squared Error: 0.23
R^2 Score: -0.11
```




```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset
data = load_iris()
X = data.data[:, :3]  # Using first 3 features
y = data.data[:, 3]   # Predicting petal width

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Train the model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate
print(f'Mean Squared Error: {mean_squared_error(y_test, y_pred):.2f}')
print(f'R^2 Score: {r2_score(y_test, y_pred):.2f}')

# Plot Actual vs Predicted
plt.figure(figsize=(8, 5))
plt.scatter(y_test, y_pred, color='purple', alpha=0.7)
plt.plot([min(y_test), max(y_test)], [min(y_test), max(y_test)], color='black', linestyle='--')
plt.xlabel('Actual Petal Width')
plt.ylabel('Predicted Petal Width')
plt.title('Multiple Linear Regression: Actual vs Predicted')
plt.grid(True)
plt.show()
```

Output:

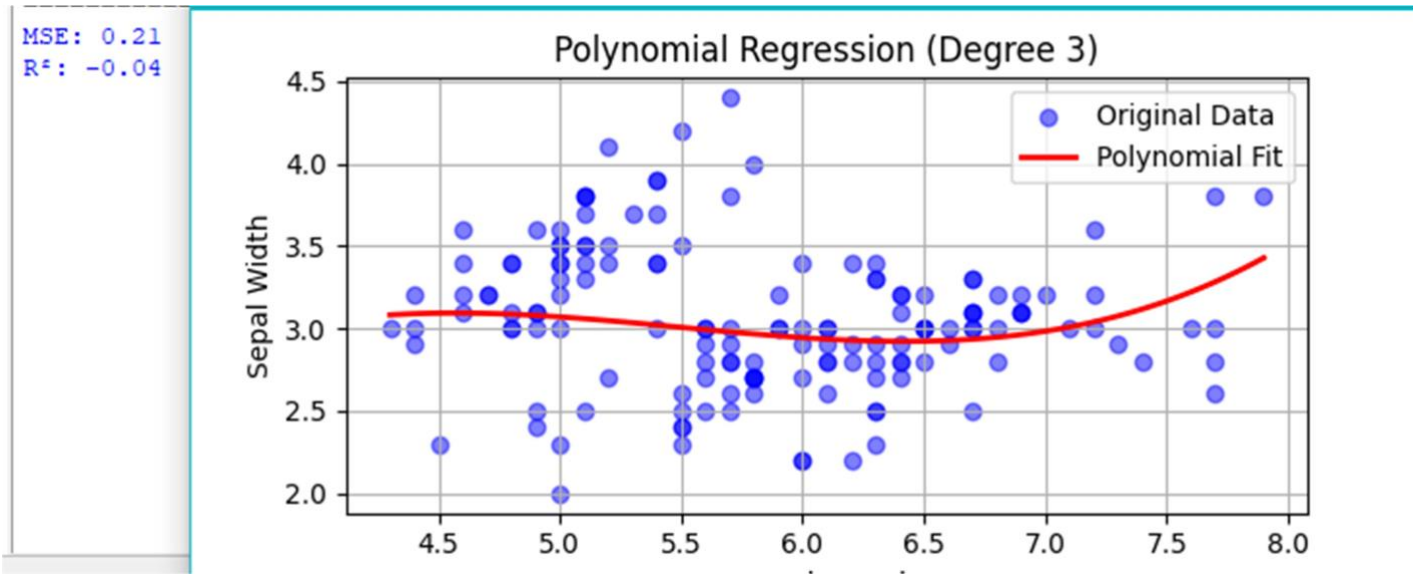


3 Polynomial Regression

```
from sklearn.datasets import load_iris
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
import matplotlib.pyplot as plt
import numpy as np

# Load dataset
data = load_iris()
X = data.data[:, 0].reshape(-1, 1)  # Use a single feature (sepal length) for simplicity
y = data.data[:, 1]  # Target: sepal width

# Generate polynomial features (degree=3)
poly = PolynomialFeatures(degree=3)
X_poly = poly.fit_transform(X)

# Train/test split
X_train, X_test, y_train, y_test = train_test_split(X_poly, y, test_size=0.3, random_state=42)

# Train the model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluation
print(f'MSE: {mean_squared_error(y_test, y_pred):.2f}')
print(f'R²: {r2_score(y_test, y_pred):.2f}')

# Plotting (on the original feature space)
X_plot = np.linspace(X.min(), X.max(), 100).reshape(-1, 1)
X_plot_poly = poly.transform(X_plot)
```

```
y_plot = model.predict(X_plot_poly)

plt.scatter(X, y, color='blue', label='Original Data', alpha=0.5)
plt.plot(X_plot, y_plot, color='red', linewidth=2, label='Polynomial Fit')
plt.xlabel('Sepal Length')
plt.ylabel('Sepal Width')
plt.title('Polynomial Regression (Degree 3)')
plt.legend()
plt.grid(True)
plt.show()
```

Output:



4 Lasso And Ridge Regression

```
from sklearn.datasets import load_iris
from sklearn.linear_model import Lasso, Ridge
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset
data = load_iris()
X = data.data[:, :3]  # Using first three features
y = data.data[:, 3]   # Predicting petal width

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Lasso Regression
lasso = Lasso(alpha=0.1)
lasso.fit(X_train, y_train)
y_pred_lasso = lasso.predict(X_test)
print('Lasso Regression:')
```

```
print(f'Mean Squared Error: {mean_squared_error(y_test, y_pred_lasso):.2f}')
print(f'R^2 Score: {r2_score(y_test, y_pred_lasso):.2f}')

# Ridge Regression
ridge = Ridge(alpha=1.0)
ridge.fit(X_train, y_train)
y_pred_ridge = ridge.predict(X_test)
print('\nRidge Regression:')
print(f'Mean Squared Error: {mean_squared_error(y_test, y_pred_ridge):.2f}')
print(f'R^2 Score: {r2_score(y_test, y_pred_ridge):.2f}')
```

Output:

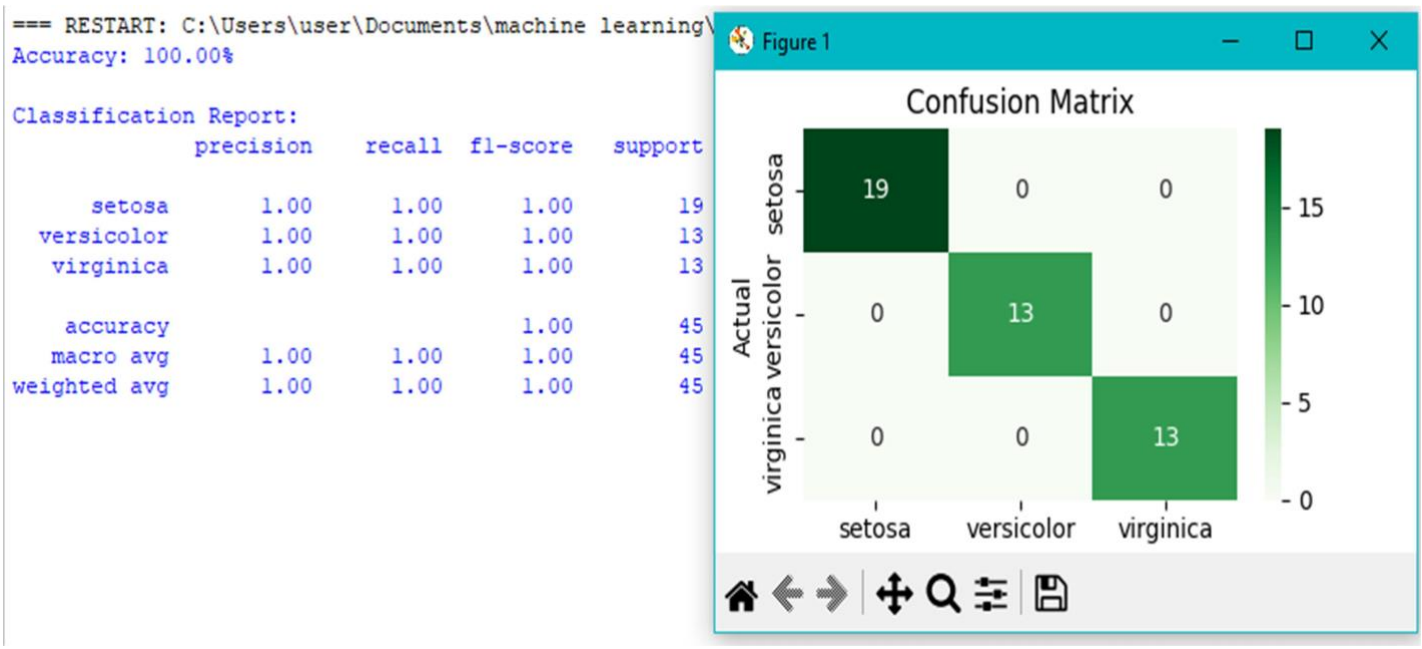
```
Lasso Regression:
Mean Squared Error: 0.04
R^2 Score: 0.94

Ridge Regression:
Mean Squared Error: 0.04
R^2 Score: 0.94
```

5 Logistic regression

```
from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
data = load_iris()
X, y = data.data, data.target

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Initialize and train model
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Evaluate
print(f'Accuracy: {accuracy_score(y_test, y_pred) * 100:.2f}%')
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=data.target_names))
```

```
# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Greens', xticklabels=data.target_names,
yticklabels=data.target_names)
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```

Output:



6 Artificial Neural Network

```
import tensorflow as tf
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report

# Load and preprocess data
X, y = load_breast_cancer(return_X_y=True)
X = StandardScaler().fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=0)

# Build ANN model
model = tf.keras.Sequential([
    tf.keras.layers.Dense(16, activation='relu'),
    tf.keras.layers.Dense(1, activation='sigmoid')
```

```
])

# Compile and train
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
model.fit(X_train, y_train, epochs=20, batch_size=16, verbose=0)

# Evaluate
y_pred = (model.predict(X_test) > 0.5).astype("int32")
print(classification_report(y_test, y_pred))
```

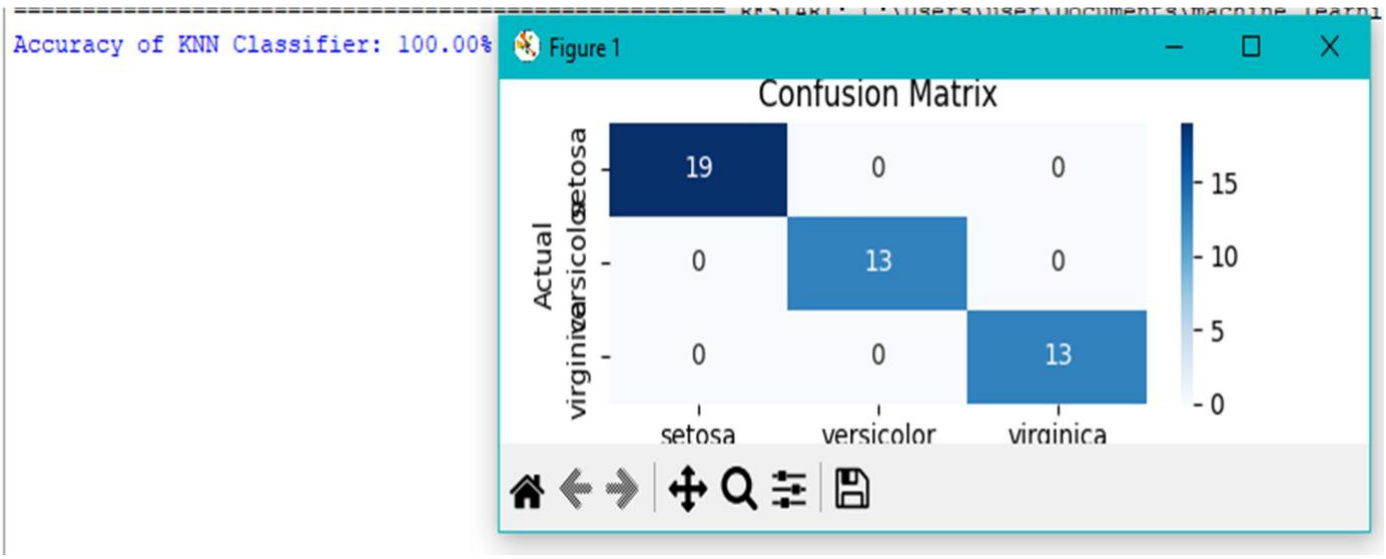
Output:

7 Knn

```
# Import necessary libraries
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.datasets import load_iris
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import matplotlib.pyplot as plt
import seaborn as sns

# Load the Iris dataset
data = load_iris()
X = data.data  # Features
y = data.target  # Target labels

# Split the data into training and testing sets (70% training, 30% testing)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Initialize the KNN model (use 5 neighbors here)
knn_model = KNeighborsClassifier(n_neighbors=5)

# Train the model using the training data
knn_model.fit(X_train, y_train)

# Predict the target labels for the test data
y_pred = knn_model.predict(X_test)

# Calculate the accuracy of the model
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy of KNN Classifier: {accuracy * 100:.2f}%')

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=data.target_names,
yticklabels=data.target_names)
```

```
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()

# Classification Report (Precision, Recall, F1-Score)
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=data.target_names))
```

Output:

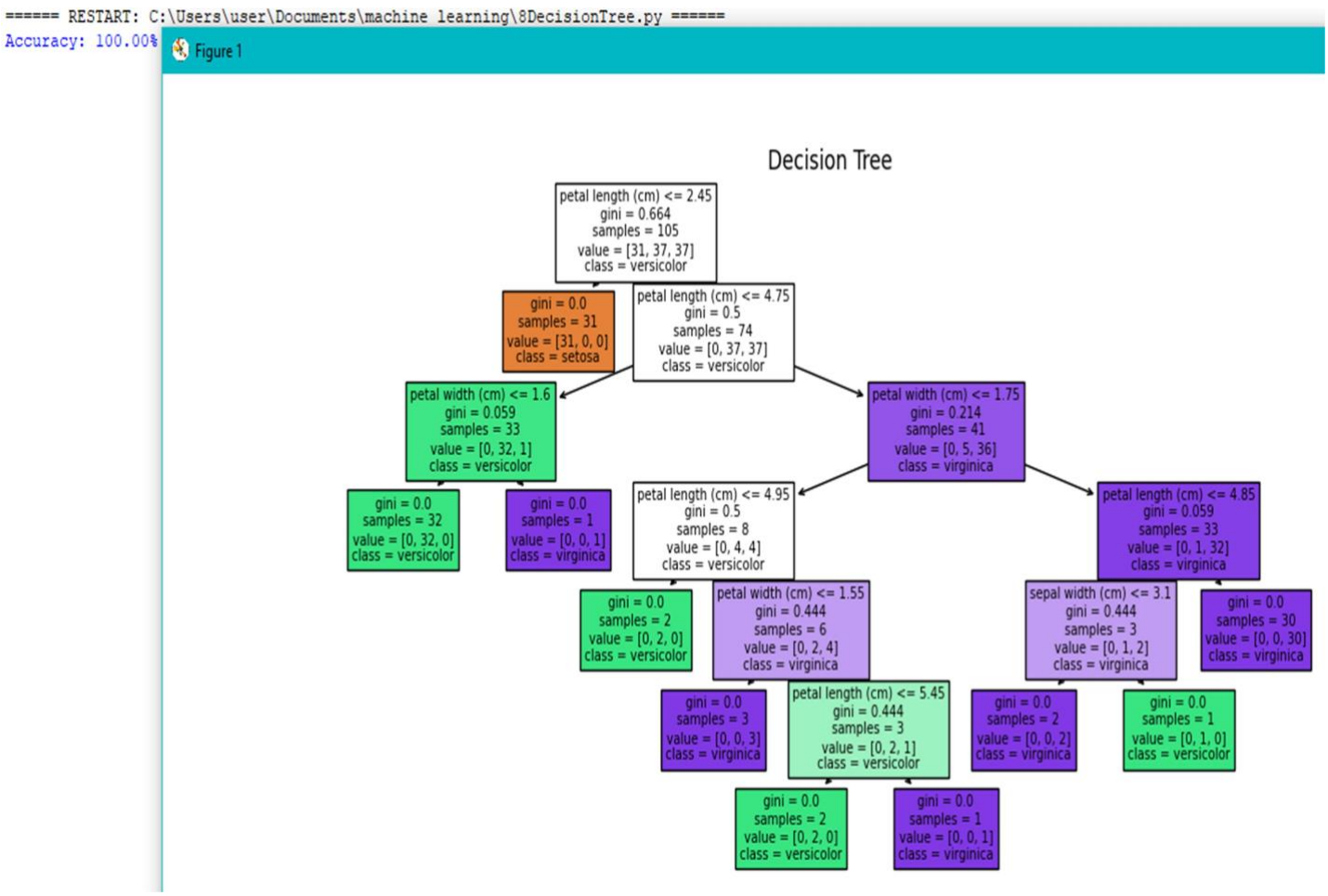


8 Decision Tree Classification

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt

# Load dataset
data = load_iris()
X, y = data.data, data.target

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Initialize and train model
model = DecisionTreeClassifier()
model.fit(X_train, y_train)
```

```
# Predict
y_pred = model.predict(X_test)

# Evaluate
print(f'Accuracy: {accuracy_score(y_test, y_pred) * 100:.2f}%')

# Plot tree
plt.figure(figsize=(12,8))
plot_tree(model, feature_names=data.feature_names, class_names=data.target_names, filled=True)
plt.title('Decision Tree')
plt.show()
```

Output:



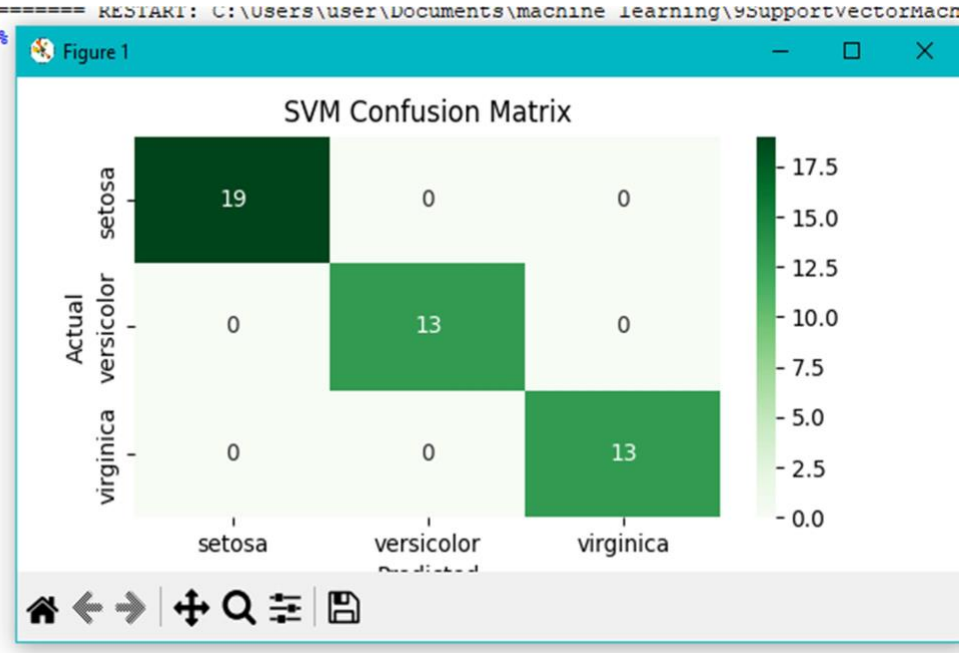
9 Support Vector Machine (SVM)

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load dataset
data = load_iris()
X = data.data  # Features
y = data.target  # Target labels

# Train/test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Initialize and train SVM model
model = SVC(kernel='linear')  # You can also try 'rbf', 'poly', or 'sigmoid'
model.fit(X_train, y_train)

# Predict on test set
y_pred = model.predict(X_test)

# Evaluate
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy of SVM Classifier: {accuracy * 100:.2f}%')

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Greens',
            xticklabels=data.target_names, yticklabels=data.target_names)
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('SVM Confusion Matrix')
plt.show()

# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=data.target_names))
```

Output:

```
========================================= RESTART: C:\Users\user\Documents\Machine learning\9SupportVectorMach
Accuracy of SVM Classifier: 100.00%
```



10 K means Clustering

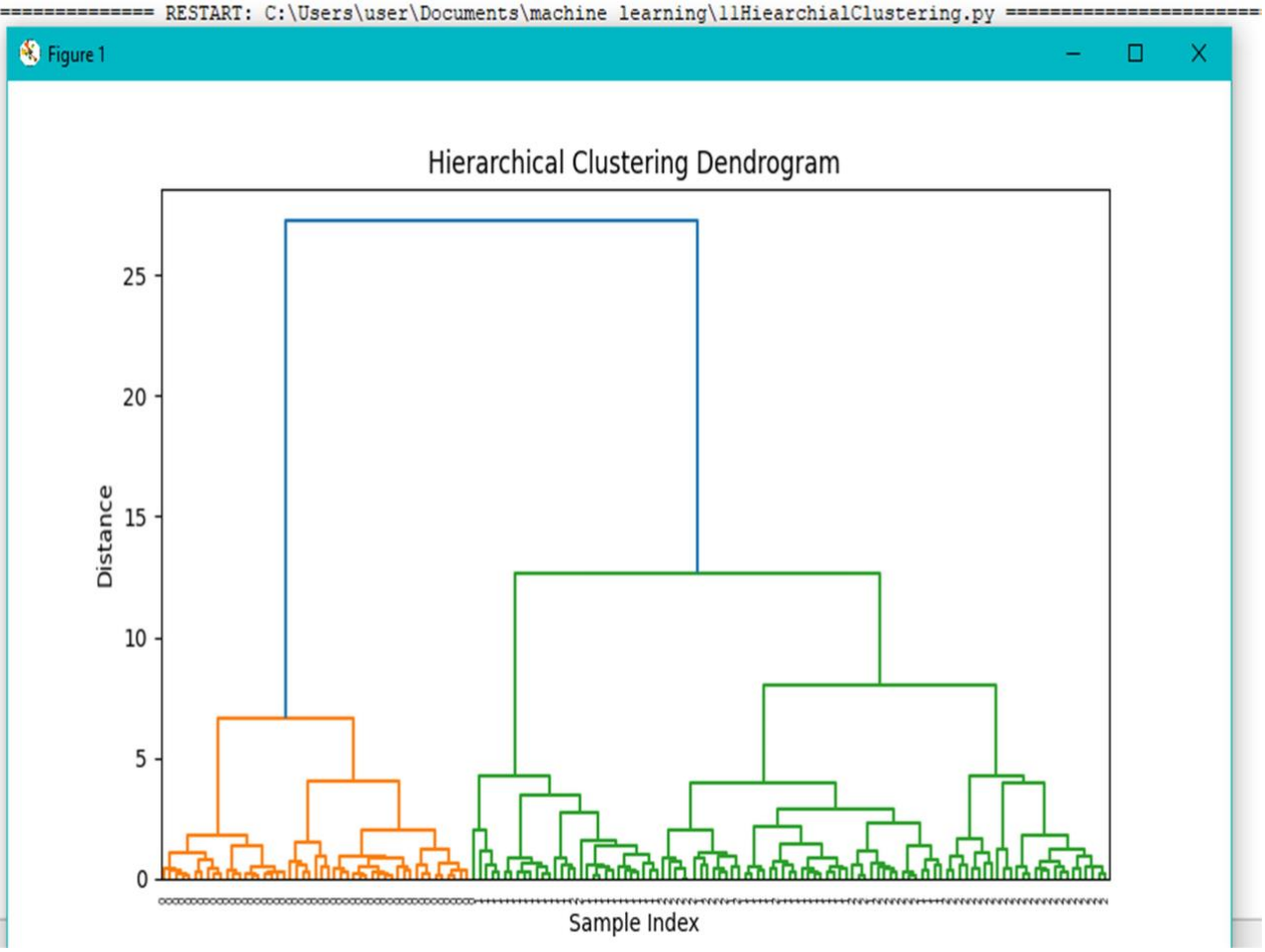
```
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# Load dataset
data = load_iris()
X = data.data

# Initialize and fit KMeans
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
y_kmeans = kmeans.fit_predict(X)

# Reduce dimensions for visualization
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

# Plot clusters
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y_kmeans, cmap='viridis', label='Cluster')
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1], c='red', marker='X', s=200,
label='Centroids')
plt.title('K-Means Clustering')
plt.xlabel('PCA 1')
plt.ylabel('PCA 2')
plt.legend()
plt.show()
```

Output:

```
============ RESTART: C:\Users\user\Documents\machine learning\10KmeansClustering.py =====
```

11 Hierarchial Clustering

```
from sklearn.datasets import load_iris
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt

# Load dataset
data = load_iris()
X = StandardScaler().fit_transform(data.data)  # Standardize features

# Compute linkage
linked = linkage(X, method='ward')

# Plot dendrogram
plt.figure(figsize=(10, 7))
dendrogram(linked, labels=data.target, leaf_rotation=90)
plt.title('Hierarchical Clustering Dendrogram')
plt.xlabel('Sample Index')
plt.ylabel('Distance')
plt.show()
```

Output:

```
================== RESTART: C:\Users\user\Documents\machine learning\11HiearchialClustering.py ========================
```